Assignment 1: Claim Report Generator

Fusion Galaxxy | Software / AI-ML Intern | about 4 hours

Why this task

Our claims system holds structured data about each repair claim. Surveyors and insurers want a clean report built from it. This tests whether you can turn messy real data into a report people can trust, in more than one format.

What is in the folder

File	What it is
CLAIM_A_data.json, CLAIM_B_data.json, CLAIM_C_data.json	Fields copied from the claims system for three real claims (redacted). Machine-parsed, so they can be wrong.
CLAIM_A/B/C estimate images	The original repair estimate each garage submitted (photo or scan). Use them to check the data. They are the source, the JSON is a copy.

What to build

1. A program that takes one claim JSON and writes a claim report in **PDF, Excel (.xlsx) and HTML**. Same content in all three, laid out for a person to read.
2. The report must show: claim summary, estimate lines (part, qty, amount, depreciation %, GST %), and totals.
3. The numbers in the files do not agree with each other. Each claim has several totals: the claims system amount, the stated estimate total, the sum of the parsed lines, and the total printed on the estimate image. Show each one with its source label. Do not pick one silently. Add a clear "Check" section that lists every mismatch and by how much.
4. Handle missing, null or odd fields without crashing. Say "not available" instead of showing 0.
5. Keep the three formats consistent. A short test that runs your generator on all three claims is a plus.

How we review

We will run your program on claims you have not seen. We look at: correct numbers, honest handling of mismatches, report readability in all three formats, error handling, and a clear README.

Rules

- The files are redacted copies of real 2025 claim records. Black boxes are on purpose. Do not try to recover or guess what is under them.
- Use the files only for this assignment. Do not share them or put them in a public place. Your repo must be private.
- AI tools are allowed. Say which ones you used and what you changed by hand.
- Do not put API keys in the repo.

Submit

- A private GitHub repo with your work and a short README (how to run, assumptions, what you would do next).

- Add GitHub user **DushyantFusion** as a collaborator, then reply to the email with the repo link.
- Optional: a 3-minute screen recording.

Deadline

6 October 2026, 6:00 PM IST. If you need more time, email us before the deadline and we will share a revised task.